

# **DVS – AI DRIVEN SMART DOCUMENT VERIFICATION & INCONSISTENCY DETECTION SYSTEM**

**SRINIDHI S**

## **Problem Statement**

The rapid growth of digital document usage in banking, education, government, and corporate sectors has increased document forgery and identity fraud. Traditional manual verification methods are slow, error-prone, and difficult to scale. Existing systems mainly rely on basic OCR and lack intelligent authenticity verification, seal validation, biometric matching, and cross-document consistency checks. There is a need for an automated AI-based system that can accurately verify documents and detect fraudulent inconsistencies in real time.

## **Proposed Solution & Mechanism**

We developed an AI-powered automated document verification system that performs intelligent verification using OCR, Computer Vision, and Deep Learning techniques.

- Uploaded documents are processed using Google Gemini Vision API for OCR-based data extraction.
- Document authenticity is analyzed using AI-based classification techniques.
- Face detection & biometric matching verify identity consistency between documents/selfies.
- Seal/stamp verification detects official seals using image analysis techniques.
- Cross-document validation compares extracted fields like name, DOB, and ID number.
- Flask-based web application provides real-time APPROVE / REJECT verification results with detailed reports.

The system achieved high verification accuracy by integrating OCR, face matching, seal verification, and cross-document validation into a unified pipeline.

## **Tech Stack Used**

- ✓ Frontend & Backend: Flask (Python), HTML, CSS, JavaScript
- ✓ OCR & AI Processing: Google Gemini Vision API
- ✓ Deep Learning & CV: OpenCV, Face Detection, Image Processing
- ✓ Data Processing: Pandas, NumPy, JSON
- ✓ Verification Techniques: OCR, Seal Verification, Face Matching, Cross Validation
- ✓ Libraries & Tools : Pillow, Werkzeug, Glob, Datetime
- ✓ Deployment & Tools : GitHub, VS Code, Flask Server

## **Demo Link (Linkedin)**

[https://www.linkedin.com/posts/ssrinidhi18112004\\_hey-tech-folks-excited-to-share-the-activity-7458407775551926273-uoMy?utm\\_source=share&utm\\_medium=member\\_desktop&rcm=ACoAAD5-vP8BKQe8faYSG7c\\_K3JlJ7Gf7ZWmTck](https://www.linkedin.com/posts/ssrinidhi18112004_hey-tech-folks-excited-to-share-the-activity-7458407775551926273-uoMy?utm_source=share&utm_medium=member_desktop&rcm=ACoAAD5-vP8BKQe8faYSG7c_K3JlJ7Gf7ZWmTck)